Finding the shortest path between two points on a curved surface can often be difficult. However, the length of a path on the surface of a cylinder is not changed if the curved surface is flattened. For the following problem, by unrolling the hollow cylinder and flattening it into a rectangle, a single reflection allows us to determine the ant's path.

Example 6.21

Consider a hollow cylindrical vessel, with circumference 24 cm and height 10 cm. An ant is located on the outside of vessel 4 cm from the bottom. There is a drop of honey at the diagrammatically opposite inside of the vessel, 3 cm from the top. (i) What is the shortest distance the ant would need to crawl to get the honey drop?. (ii) Equation of the path traced out by the ant. (iii) Where the ant enter in to the cylinder?. Here is a picture that illustrates the position of the ant and the honey.

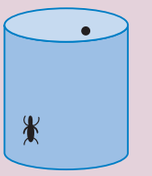


Figure 6.31

Solution:

By unrolling the hollow cylinder and flattening it into a rectangle, and with a single reflection allows us to determine the ant's path, as shown the figure. Let the base line x-axis in cm. and the vertical line through A (initial position of the ant) be the y-axis. Let H be the position of honey drop and E be the entry point of ant inside the vessel. From the given information we have

Let $A(x_1, y_1)$ and $H(x_2, y_2)$ be (0,4) and (12,13) respectively

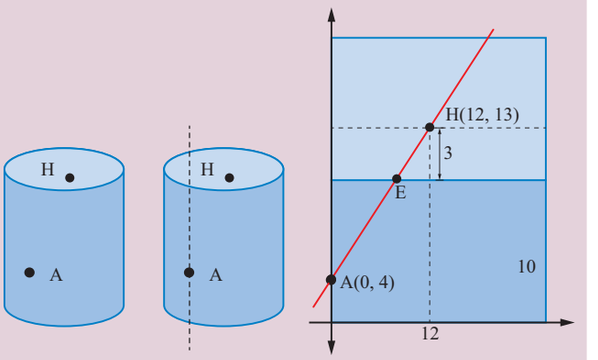


Figure 6.32

(i) The shortest distance between A and H is

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{12^2 + 9^2} = 15$$

the ant would need to crawl 15 cm to get the honey.

(ii) The equation of the path *AH* is $\dfrac{y-4}{13-4} = \dfrac{x-0}{12-0}$

$$y = 0.75x + 4 \tag{6.19}$$

(iii) At the entry point E, $y = 10 \Rightarrow x = 8$

$$E = (8,\ 10)$$

Taking the orign at different location, different form of equation can be obtained, but the path and distance are the same as above.